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## Industrial IoT and Industry 4.0

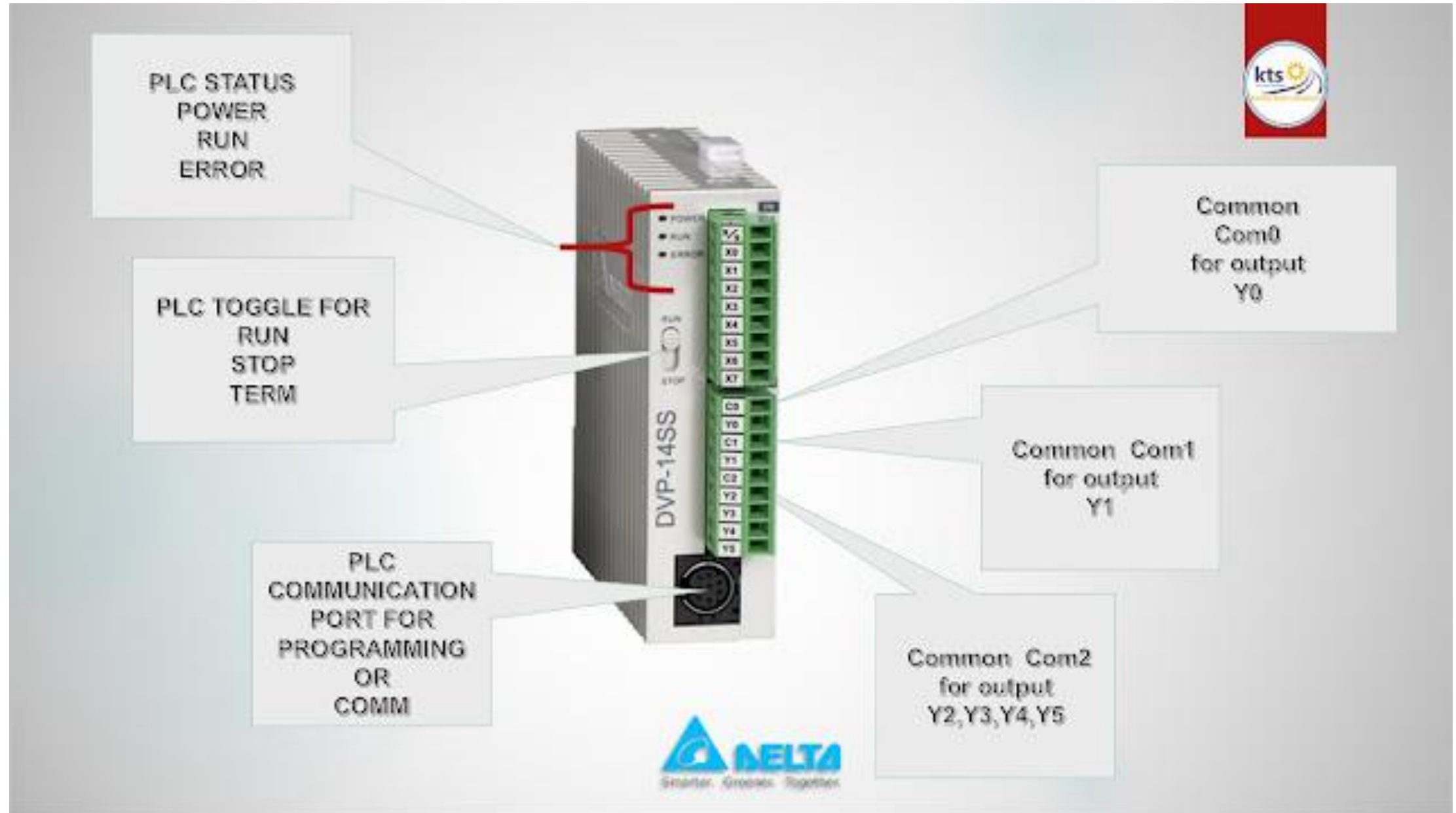
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# **DVP14SS2 Delta PLC**

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- The **DVP14SS2 Delta PLC** is a high-performance, compact programmable logic controller designed for efficient industrial automation. It offers fast processing speed, reliable performance, and flexible control functions, making it ideal for various automation applications such as machinery control, production lines, and process monitoring.
- This PLC features **8 digital inputs and 6 digital outputs**, supporting both sink and source wiring configurations. It provides **stable communication, easy programming, and expandability** through multiple I/O modules and communication ports. The DVP14SS2 supports **manual and automatic operation modes**, allowing easy integration and control in industrial environments.
- Engineered for durability and ease of maintenance, the DVP14SS2 operates on **24V DC power** and ensures long-term reliability. It is a trusted choice across industries for **cost-effective and efficient automation solutions**.

# Key Specifications & Features

| Feature                     | Description / Spec                                                                                |
|-----------------------------|---------------------------------------------------------------------------------------------------|
| Base I/O (MPU points)       | 14 total (8 DI + 6 DO)                                                                            |
| Max I/O Expandability       | Up to 494 I/O points (14 + 480)                                                                   |
| Program Capacity            | 8,000 steps (8k)                                                                                  |
| CPU / Processing            | 32-bit CPU (for high-speed processing)                                                            |
| Instruction Execution Speed | Typical basic instruction “LD” ~ 0.35 μs                                                          |
| Communication Interfaces    | Built-in RS-232 & RS-485 ports; supports Modbus ASCII/RTU; can act as master or slave             |
| High-Speed Pulse Outputs    | 4 independent pulse outputs (Y0 to Y3), up to 10 kHz                                              |
| High-Speed Counters         | Built-in high-speed counter support                                                               |
| PID Auto-Tuning Support     | The PLC supports PID auto-tuning and can save parameter values automatically once tuning is done. |
| Data Memory / Registers     | (From some sources) Data registers ~5,000 words                                                   |
| Output Types / Variants     | e.g. DVP14SS211R (relay output), DVP14SS211T (transistor output)                                  |
| Power Supply                | Typically works with 24 V DC (for the slim series)                                                |
| Mounting Style              | DIN rail mounting (common for slim PLCs)                                                          |
| Certifications / Compliance | CE, cULus (in many regions)                                                                       |

# Power Terminals (Main Terminal Block)

- The main terminal block contains the power connection points for the PLC's Central Processing Unit (CPU).

| Terminal | Function                 | Detail                                                                                                     |
|----------|--------------------------|------------------------------------------------------------------------------------------------------------|
| 24VDC    | Power Input (+)          | Connects to the Positive lead of the 24V DC power supply. The DVP-SS2 series operates on DC 24V(−15%~20%). |
| 0V       | Power Input (-) / Ground | Connects to the Negative lead of the 24V DC power supply and often serves as the system ground reference.  |

# Key Components and Ports

| Feature                            | Description                                    | Function                                                                                                                                                                                                            |
|------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model Label                        | DVP-14SS2                                      | Confirms the unit is a 14-point MPU (8 inputs, 6 outputs).                                                                                                                                                          |
| Green Terminal Blocks              | I/O Terminals                                  | These removable blocks are where you connect the field devices. The terminals are used for the 8 Digital Inputs (X0-X7) and 6 Digital Outputs (Y0-Y5), along with the 24VDC power supply connection.                |
| Circular Port                      | RS-232 Communication Port (MINI-DIN 8-pin)     | This is the primary port for programming and monitoring the PLC from a PC using Delta's ISPSoft software. It also supports Modbus communication.                                                                    |
| Small White/Green Port (Left Side) | Expansion Port                                 | Used to connect expansion modules (digital I/O, analog I/O, temperature modules) to extend the PLC's capabilities up to 494 total I/O points.                                                                       |
| Slider Switch (Top-Right)          | RUN/STOP Switch                                | A physical switch to change the PLC's operational mode. RUN executes the program, while STOP halts program execution for editing or uploading.                                                                      |
| Indicator Lights (Far Right)       | LED Indicators (POWER, RUN, ERROR, I/O Status) | Provide immediate visual feedback on the PLC's status. POWER indicates the unit is receiving power, RUN/ERROR indicates the CPU's current operating status, and others show the ON/OFF state of the X and Y points. |
| Finned Casing                      | Heat Dissipation                               | The grooves or "fins" are part of the heat sink design to help dissipate heat from the internal electronics, essential for reliable operation in industrial environments.                                           |



# Key Components and Ports

| Port           | Type/Connector                                                          | Function                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COM1 (RS-232)  | 8-pin Mini-DIN (Circular)                                               | This is the primary port for programming and monitoring. It connects directly to a PC or programming panel for uploading/downloading programs and debugging. It supports the Modbus ASCII/RTU protocol.                                                                                                                                                                                                           |
| COM2 (RS-485)  | Screw Terminals (on the same block as I/O, typically labeled D+ and D-) | Used for network communication with other devices like Human-Machine Interfaces (HMIs), Variable Frequency Drives (VFDs), or other PLCs. It supports the Modbus ASCII/RTU protocol in master or slave modes. (Note: Some very compact 10/14-point SS2 models may not have the physical RS-485 terminals, relying only on RS-232 or the expansion port for networking, but the DVP-14SS2 typically includes both). |
| Expansion Port | IDC or Small White Connector (Left Side)                                | Used to connect DVP-S series expansion modules, such as digital I/O, analog I/O, or temperature modules, to increase the total I/O capacity.                                                                                                                                                                                                                                                                      |

# I/O Terminals (Main Terminal Blocks)



## Digital Inputs (DI)

| Category         | Specification  | Detail                                                     |
|------------------|----------------|------------------------------------------------------------|
| Total Points     | 8 Inputs       | X0 to X7                                                   |
| Function         | Standard Input | Connects to external devices (switches, buttons, sensors). |
| High-Speed X0-X3 | 20kHz Max.     | For high-speed counting (e.g., encoders).                  |
| High-Speed X4-X7 | 10kHz Max.     | For auxiliary high-speed counting.                         |
| Wiring Type      | Sink/Source    | SINK (common tied to 24VDC) or SOURCE (common tied to 0V). |

# I/O Terminals (Main Terminal Blocks)



## Digital Outputs (DO)

| Category              | Specification          | Detail                                                                                 |
|-----------------------|------------------------|----------------------------------------------------------------------------------------|
| Total Points          | 6 Outputs              | Y0 to Y5                                                                               |
| Output Type           | Model Suffix R         | Relay output (used for general purpose switching of AC/DC loads).                      |
| Output Type           | Model Suffix T         | Transistor (NPN/PNP) output (used for high-speed switching).                           |
| High-Speed Capability | 4 points of 10kHz      | Y0 to Y3 on Transistor models for pulse output (motion control).                       |
| Wiring Requirement    | Common Terminals (COM) | Outputs are grouped by COM terminals, which must be wired to the correct load voltage. |

## Working Principle

A **PLC (Programmable Logic Controller)** is a microprocessor-based device designed to control machinery or processes automatically.

The DVP14SS2 operates by continuously performing the following **control cycle**:

### Input Scan:

- Reads the status of all input devices (sensors, switches, etc.) connected to the input terminals.

### Program Execution:

- Executes the user-written logic program stored in its memory. This logic is created using **ladder diagrams, function block, or structured text** in WPLSoft.

## Output Scan:

- Updates the status of all output devices (motors, relays, actuators, etc.) according to the executed logic.

## Housekeeping:

- Performs internal diagnostics, communication handling, and updates the system status.
- This process repeats cyclically within milliseconds, allowing real-time and deterministic control of industrial equipment.

The DVP14SS2 operates on the standard **cyclic scan** system, but its performance is highly optimized by its 32-bit CPU:

- **Input Scan/Sampling:**
  - The PLC reads the state of all physical inputs (from X0 to X7 and expansion modules) and updates the **Input Image Register** in the PLC's memory.
- **Program Execution:**
  - The CPU executes the user program (Ladder Logic) sequentially from the first line to the END instruction.
  - It uses the data stored in the Input Image Register and the internal memory devices.

- **Output Refresh:**

- The execution results are written to the **Output Image Register**, which is then used to refresh (turn ON/OFF) the physical output points (Y0 to Y5 and expansion modules).

The total time for these three steps is the **scan time**. Since the basic instruction speed is  $0.35 \mu\text{s}$ , a small program can complete a scan in less than 1 millisecond, which is why it's considered a fast-response controller.

# High-Speed Control Architecture

The DVP14SS2 is a compact motion control device designed for applications requiring precise speed and position control.

## High-Speed Pulse Output

- **Outputs:** Supports **4 independent channels** (Y0, Y1, Y2, Y3).
- **Frequency:** Max. **10kHz** per channel.
- **Functions:** Used for controlling stepper or servo motor drivers via pulse trains (PLSY) or specific motion instructions like DDRVI (positioning via absolute/incremental mode). The high frequency is essential for controlling the speed and position of the motor.



# High-Speed Counters (HSC)

- **Inputs:** Inputs X0 to X7 can be configured for high-speed counting, bypassing the main PLC scan for rapid response.
- **Max Frequency:** Inputs X0 to X3 support a max frequency of **20kHz**, while X4 to X7 support **10kHz**.
- **Application:** Used to capture pulses from sensors like encoders. For example, X0 and X1 can be configured for 2-phase A/B counting (quadrature) to track both position and direction.

# Memory Map and Register Types

The PLC's memory is organized into various devices (registers) used to store bits and data words.

| Device Type                         | Address Range (Octal) | Size (Points/Words) | Function & Latched Status                                                                                                                                                            |
|-------------------------------------|-----------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs                              | X0 to X377            | 256 Points Max.     | Physical Input Status (Read-Only).                                                                                                                                                   |
| Outputs                             | Y0 to Y377            | 256 Points Max.     | Physical Output Status (Write-Only).                                                                                                                                                 |
| Auxiliary Relays<br>(Internal Bits) | M0 to M1999           | Approx. 2000 Bits   | General purpose internal coils/flags. A large portion is Non-Latched (cleared on power cycle), while higher-numbered M registers are often Latched (retain status after power loss). |
| Timers                              | T0 to T255            | 256 Timers          | Timers and their current values. Resolution is typically 1ms or 100ms.                                                                                                               |
| Counters                            | C0 to C255            | 256 Counters        | Counters and their current values.                                                                                                                                                   |

# Functional Capabilities and Applications

Beyond the basic specs, the DVP14SS2 offers several features that make it suitable for a range of industrial applications.

- **Motion Control:**

- The PLC supports 4 independent points of high-speed pulse output (Y0-Y3) up to 10kHz.
- This allows for direct control of devices like stepper or servo drives, which is essential in applications such as **winding machines for tension control** or simple positioning systems.

- **Process Control:**

- With built-in **PID auto-tuning** functionality, this PLC can automatically calculate and save optimal PID parameters.
- This makes it well-suited for controlling continuous process variables like **temperature, pressure, or flow rate** in industrial settings.

- **Expansion Capabilities:**

- While the base unit is compact, the system can be expanded with a variety of digital, analog, and special function modules (like temperature measurement PT100/thermocouple or additional communication modules) to achieve up to 494 I/O points.
- This flexibility allows it to adapt to more complex machine requirements.

- **Network Communication:**

- A common and powerful application is using the built-in **RS-485 port and Modbus protocol** to create a network.
- The DVP14SS2 can act as a **Modbus Master** to control other devices like Variable Frequency Drives (VFDs).
- For example, it can send commands over Modbus to start/stop a motor and precisely control its speed by writing to the VFD's holding registers

# Features

- High-speed and stable processing for time-critical operations.
- Compact and lightweight design suitable for control panels.
- Supports **PID control**, **PWM output**, and **high-speed counters**.
- Integrated communication ports for **Modbus networking**.
- Easy-to-use **WPLSoft** interface for programming and debugging.
- Expandable I/O for modular system design.
- Non-volatile memory for program retention during power loss.

# Applications

The DVP14SS2 is used in a wide variety of **industrial automation** and **process control** systems, such as:

- Conveyor belt control systems
- Packaging and labeling machines
- Automatic door systems
- Material handling systems
- Pump and motor control
- Assembly line automation
- Temperature and lighting control systems

# Advantages

- Fast instruction execution speed
- Reliable and long operational life
- Flexible communication with other PLCs and HMIs
- Cost-effective solution for small automation systems
- Simple installation and maintenance

## Limitations / Cautions

- The **I/O count** is limited; for large-scale systems, you'll need more powerful or modular PLCs.
- The **program memory (8k steps)** is modest — complex logic or large routines might push limits.
- **Speed** is adequate for many tasks, but not for ultra-high-speed real-time control (e.g. in precise motion control demanding  $\mu$ s-level loops).
- Ensure **proper isolation, noise filtering**, and wiring practices especially in industrial environments.
- Choose the correct **output variant** (relay vs transistor) based on the loads you need to switch.
- For **expansion modules**, make sure the compatibility and communication protocols are matched.



# Programming and Software

The DVP14SS2 is programmed using **Delta's WPLSoft** software.

Key programming methods include:

- **Ladder Diagram (LD):** Most commonly used for logic and control.
- **Instruction List (IL):** Used for complex calculations.
- **Function Blocks (FB):** For PID, counters, timers, and motion functions.

Programs can be downloaded to the PLC via USB or RS-232 port.

# Ladder Diagram (LD)

- A Ladder Diagram (LD) for the Delta DVP-14SS2 PLC is a graphical programming method used for control logic, especially in industrial automation.
- It visually represents the sequence of switching operations, using symbols for contacts (inputs), coils (outputs), timers, and more, arranged like electrical relay circuits.

## In Ladder Diagram:

- The **left vertical line** represents the **power supply (logic true rail)**.
- The **right vertical line** represents the **ground (logic false rail)**.
- **Horizontal lines** between them are called **rungs**, each representing a **logical operation or control task**.

# Structure

## Power Rails:

- The two vertical lines (representing power/voltage) that run down the side of the diagram. The left rail is often called the "**Source**" or "**Hot**" rail, and the right rail is the "**Return**" or "**Neutral**" rail.

## Rung:

- Each horizontal line of logic, resembling a rung on a ladder. A rung must begin at the left rail and typically ends with an output instruction (or "coil") on the right.

## Logic Flow:

- Power flows from left to right. A rung is considered logically "True" (energized) if there is a complete path (continuity) from the left rail to the output instruction.

# Ladder Logic Symbols Used in Delta PLC

| Symbol | Name                         | Description / Function                                         |
|--------|------------------------------|----------------------------------------------------------------|
| [ ]    | Normally Open Contact (NO)   | Turns ON when the associated input is ON.                      |
| [ \ ]  | Normally Closed Contact (NC) | Turns ON when the associated input is OFF.                     |
| ( )    | Coil (Output)                | Represents an output device like a motor, lamp, or relay coil. |
| [SET]  | Set Coil                     | Turns ON an output latch (remains ON until reset).             |
| [RST]  | Reset Coil                   | Turns OFF a previously latched output.                         |
| [TMR]  | Timer                        | Activates an output after a preset time delay.                 |
| [CNT]  | Counter                      | Counts input pulses up to a preset number.                     |

# Logic Execution Sequence

- **Input Scan:** The PLC checks the status (ON/OFF) of all input devices connected to input terminals (e.g., sensors, switches).
- **Program Execution:** The ladder logic program is executed from top to bottom, left to right.
- **Output Update:** Based on logic results, the PLC updates the outputs (e.g., motors, relays, lamps).
- **Repeat Cycle:** This scanning process repeats continuously in milliseconds.

# Ladder Logic for the Delta DVP14SS2

When you program the **Delta DVP14SS2** in ISPSoft or WPLSoft:

- **Inputs (X):** X0, X1, X2, ... (e.g., connected to sensors, buttons)
- **Outputs (Y):** Y0, Y1, Y2, ... (e.g., connected to motor starters, solenoids, lights)
- **Auxiliary Relays (M):** M0, M1, M2, ... (internal flags for intermediate logic)
- **Timers (T):** T0, T1, T2, ... (e.g., T0 K50 for a 5-second delay, where K50 = 50 x 100ms)
- **Counters (C):** C0, C1, C2, ...
- **Data Registers (D):** D0, D1, D2, ... (used for storing numerical values)

A simple program for the DVP14SS2 to turn on an output (Y0) when an input (X0) is active would look like this in the software:

# Ladder Logic is so popular

## Ease of Understanding

- Ladder Logic is visually based on **electrical relay diagrams**.
- Engineers familiar with **relay-based control systems** could transition to PLCs without learning complex programming languages.
- The “rungs” of a ladder are intuitive: **inputs → logic → outputs**, similar to how circuits work.

## Visual Representation

- Each rung represents a logic operation in a **graphical, block-like form**.
- It's easier to **diagnose problems** by looking at the ladder diagram, compared to textual code.
- Maintenance and troubleshooting are faster on-site.

## Standardization

- Ladder Logic is standardized under **IEC 61131-3**, making it compatible across PLC brands.
- This allowed companies to train engineers on a **common programming method**.

## Real-Time Control

- PLCs execute Ladder Logic in **scans**, checking input conditions and updating outputs in real time.
- This makes it ideal for **industrial processes** where timing and sequencing matter.

## Modularity

- Functions like **timers, counters, and PID loops** can be implemented as **Function Blocks** in ladder logic.
- Complex control can be built **step by step**, rung by rung.



## Low Learning Curve for Industrial Applications

- No need to know traditional programming (C, Python).
- Electrical engineers can implement control logic **directly**.
- Documentation and collaboration are simplified because the diagram looks like wiring schematics.

## Industry Adoption

- Since PLCs became standard in factories, **Ladder Logic became the “default” language**.
- Over decades, a large workforce became proficient in it, reinforcing its popularity.

# Example Ladder Program

## Lamp Control with LDR Sensor

- Turn ON a lamp (Y0) automatically when light intensity (LDR sensor at X0) is low.

|----[ /X0 ]------( Y0 )----|

### Explanation:

- X0 is ON when light intensity is high (LDR active).
- When the input X0 is OFF (low light), Y0 turns ON (lamp glows).

# Advantages of Ladder Diagram

- Easy to read, write, and troubleshoot
- Resembles traditional electrical wiring diagrams
- Well supported by all major PLCs, including Delta DVP series
- Ideal for logic-based automation tasks

# Applications

- Motor start/stop and interlocking
- Conveyor control systems
- Level and temperature control
- Automatic lighting systems
- Sequential process control

# Instruction List (IL)

- **Instruction List (IL)** is a **low-level, text-based programming language** used in PLCs (Programmable Logic Controllers) such as the **Delta DVP14SS2**.
- It is similar to assembly language and provides a compact, efficient way to write control logic for industrial automation tasks.
- Each line in an IL program represents a single operation, such as loading a value, performing a logical operation, or writing to an output.

## Basic Concept

The IL language uses **mnemonics** (short text codes) to represent instructions.

Each instruction performs a specific action such as:

- **LD (Load):** Load an input or internal relay.
- **AND / OR:** Perform logical AND or OR operations.
- **OUT:** Send the result to an output device (like a relay or lamp).

The program executes **sequentially**, one instruction at a time, from top to bottom, making IL easy to follow for small control tasks or arithmetic operations.

# Common IL Instructions in Delta DVP14SS2

| Instruction | Description                        | Example                                  |
|-------------|------------------------------------|------------------------------------------|
| LD X0       | Load input X0 into the accumulator | Loads the status of input X0             |
| AND X1      | Logical AND with input X1          | Output is ON only if X0 and X1 are ON    |
| OR X2       | Logical OR with input X2           | Output is ON if X0 or X2 is ON           |
| OUT Y0      | Output result to Y0                | Sends the logic result to output coil Y0 |
| LDI X3      | Load inverted input X3             | Loads the NOT of X3                      |
| ANI X4      | AND NOT with X4                    | AND operation with inverted X4           |
| ORI X5      | OR NOT with X5                     | OR operation with inverted X5            |
| SET Y1      | Latch output Y1 ON                 | Keeps Y1 ON until a reset occurs         |
| RST Y1      | Reset output Y1                    | Turns OFF the latched output Y1          |

## Features of Instruction List (IL)

- **Text-based:** Uses a simple, line-by-line command structure.
- **Compact and fast:** Efficient for small programs with limited logic.
- **Precise control:** Allows direct manipulation of bits, words, and registers.
- **Arithmetic and logical operations:** Supports addition, subtraction, timers, counters, and comparison.
- **Easily transferable:** The same code can be reused across different PLC models with minor changes.



## Advantages

- Easy to **edit and debug** due to its step-by-step logic structure.
- Consumes **less memory** compared to graphical programs like Ladder Diagram.
- Suitable for **mathematical or algorithmic control** applications.
- Ideal for programmers familiar with **assembly-style coding**.

## Disadvantages

- **Less visual** than Ladder Diagram (LD), making it harder for beginners to interpret.
- **Limited readability** for large or complex control systems.
- Requires good understanding of **PLC addressing and logic flow**.

## Application in Delta DVP14SS2

In the **DVP14SS2 PLC**, the IL programming method is often used for:

- Implementing **logical control** and **arithmetic operations**.
- Handling **PID control**, **counters**, **timers**, and **comparison logic**.
- Creating compact programs where memory efficiency and execution speed are important.

## Example IL Program

```
LD  X0  
AND X1  
OUT Y0
```

### Explanation:

- Load the input **X0**.
- Perform a logical **AND** with input **X1**.
- Output the result to **Y0**.
- In this case, output **Y0** will turn ON only when both inputs **X0** and **X1** are ON.

## Function Block (FB)

A **Function Block (FB)** is a **predefined, reusable program module** that performs a specific function in PLC programming. Unlike simple ladder logic instructions, a function block can **encapsulate multiple operations** such as timers, counters, PID control, arithmetic operations, or motion control.

- Think of it as a “**black box**”: you give it inputs, it processes them according to internal logic, and it gives outputs.
- FBs can be used **multiple times** with different input/output variables without rewriting the internal logic.

# Structure of a Function Block

A typical Function Block has three main components:

## Inputs (IN):

- Variables that the FB requires to work.
- Example: start signal, sensor value, setpoint, etc.

## Outputs (OUT):

- Variables that the FB generates after processing.
- Example: motor ON/OFF, valve open/close, calculated output, etc.

## Internal/Temporary Variables (VAR/VAR\_TEMP):

- Used inside the FB for intermediate calculations.
- Not visible outside unless explicitly mapped.

## Example of FB in Delta PLC:

| FB Name | Inputs         | Outputs | Description                      |
|---------|----------------|---------|----------------------------------|
| CTU     | CU, R, PV      | Q, CV   | Up-counter: counts pulses        |
| TON     | IN, PT         | Q, ET   | On-delay timer                   |
| PID     | PV, SP, Kp, Ki | MV      | Proportional-Integral-Derivative |

# Advantages of Function Blocks

- **Reusability:** Can be used multiple times in different parts of the program.
- **Modularity:** Makes complex programs easier to understand and maintain.
- **Abstraction:** Hides internal logic, only inputs and outputs are needed.
- **Standardization:** Common FBs (Timers, Counters, PID) are standardized across PLCs.

# Examples of Common Function Blocks

## Timer FBs

- **TON (On-Delay Timer):** Output turns ON after a set delay.
- **TOF (Off-Delay Timer):** Output turns OFF after a set delay.

## Counter FBs

- **CTU (Up Counter):** Counts up to a preset value.
- **CTD (Down Counter):** Counts down from a preset value.

## Math/Logic FBs

- Arithmetic operations, comparison blocks, and logical operations can also be implemented as FBs.

## PID FB

- Used for process control like temperature, speed, and pressure.



## How to Use an FB in Ladder Diagram (LD)

- **Insert FB into a network.**
- **Assign input variables** (from sensors, switches, etc.).
- **Map output variables** (to actuators, lamps, motors, etc.).
- **Activate the FB** by enabling its rung in the ladder logic.

### Example: TON Timer in LD

|----[Start Button]----(TON Timer)----| IN: Start PT: 5s Q : Lamp

- Pressing the start button triggers the timer.
- After 5 seconds, the lamp turns ON.

# Thank You